

Supplementary Text S4. Isolation and quantification of MmuPV1 virions from papillomas

Introduction:

In this section we describe our method of isolation and quantification of MmuPV1 virions from papillomas arising in nude mice that can be serially passaged or used to infect other strains of mice. In our studies we have used both crude virus extracts or purified virus particles. Crude virus extract refers to virus present in supernatants obtained by clarifying papillomatous tissue extracts from nude mice (described in section 1.1). Our method of generation of crude virus extract has been modified from a previous protocol used to isolate merkel cell polyomavirus from human skin swabs(1). Purified virus particles refer to crude extract clarified over an optiprep gradient. Optiprep gradients have been used to purify papillomavirus generated *in vitro* (2) and have been described in detail on Dr. Chris Buck's website: https://home.ccr.cancer.gov/LCO/pseudovirusproduction.htm#_ENREF_7. We measure the amount of encapsidated viral genomic DNA to determine the concentration of "viral genome equivalents" (VGE) in any given stock of virus. For this we use a standard agarose gel electrophoresis followed by SYBR-green staining. The same gel can also be used for southern analysis using MmuPV1-specific probes (3) by nick translation as we have described in detail previously(4). Since virus yields from papillomas of nude mice are very high, we have found agarose gel electrophoresis to be sensitive enough to accurately determine virus concentrations. Other methods of virus quantification have also been described in the literature(5, 6).

Materials:

Special equipment: Homogenizer (Fisher Scientific, PowerGen 125), 5ml polyallomer tubes (Beckman #326819), Biopsy needle (SurgiPro #SPTB-014), 3ml syringe (BD Biosciences), Ultracentrifuge for >50000 rpm, Swinging Bucket Ultracentrifuge rotor (Beckman, SW55Ti), siliconized eppendorf (Fisher #05-541-63), clamp stand, gram scale

List of reagents: DPBS (Thermo Fisher Scientific # 14190250), Triton-X-100 (Pierce #28314), Benzonase (Sigma, #E1014), Collagenase H (Sigma, #11074032001), Sybr Green Stain (Thermo Fisher Scientific # S7563), 60% Idodixanol/Optiprep solution (Sigma #D1556).

Detailed protocol:

1.1 Harvesting of virus particles from papillomas

- i. Euthanize the animal from which tissue has to be harvested by CO₂ exposure as per IACUC guidelines.
- ii. Use a scalpel to excise the papilloma from the ear taking care to not collect the surrounding epithelia (non-papilloma tissue).
- iii. Weigh the papilloma and place in a 15ml round bottom falcon tube. For transportation tube is kept on ice till ready for processing at lab bench
- iv. For every 10mg wart 700µl DPBS is added to the tube
- v. Papilloma is homogenized for less than one minute using homogenizer (model no.) and suspension is immediately transferred to a siliconized 1.5ml sterile eppendorf tube.
- vi. In a 50ml falcon tube add about 25ml DPBS and wash homogenizer to collect any debris that might have got stuck in the homogenizer blades. Centrifuge falcon tube to collect

pellet and aspirate the supernatant. Resuspend pellet in 42.5µl DPBS and combine with above fraction.

vii. Add 7.5µl of Triton-X-100 solution (1% Triton-X-100).

viii. Add 2µl Benzonase endonuclease and incubate at 37°C for 20 minutes inverting tube every 5 minutes.

ix. Add 2-3mg of Collagenase H and vortex sample and incubate at 4°C overnight.

x. Adjust NaCl concentration of suspension to 0.8M by adding appropriate amount of 5M NaCl (Since suspension is 750µl, add 127.5µl of 5M NaCl).

xi. Centrifuge tube for 5 mins at 5000 x g. Transfer supernatant to fresh siliconized tube.

Collect a 20µl aliquot of the virus in a separate tube for quantification.

xii. At this point you can proceed to use crude virus extract for serial infections or purify over an optiprep gradient. Virus extract can be stored at -80°C until future use.

1.2. Purification of virus extract over optiprep gradient

xiii. At this point you can proceed to use crude virus extract for serial infections or purify over an optiprep gradient.

xiv. At least 2 hours before spinning, prepare a three-step optiprep gradient (27%-33%-39%) and let it rest. First pipette 1.5ml of 27% optiprep solution directly to the 5ml centrifuge tube. Now using a long bone-marrow biopsy long needle, dispense 1.5ml 33% optiprep solution slowly at the bottom of the tube. If layered correctly you will see three separate layers of the two solutions. Now dispense 1.5ml of 39% optiprep solution at the bottom of the tube that will result in three steps of the gradient. Let the gradient stand at room temperature for 2-4 hours.

xv. Gently layer crude lysate on top of optiprep gradient and spin at 16°C, at 50000 RPM for 3.5 hours using the SW-55 swinging bucket rotor. Prior to loading the tubes balance the tubes using DPBS by weighing using a gram scale.

- xvi. After centrifugation carefully remove the 5ml tube using forceps and mount on a clamp stand. With the help of a 27-gauge needle carefully make a hole at the bottom of the tube slightly off center. Now collect virus fractions in a drop-wise manner as follows: collect fractions 1 and 2 as approximately 500µl, fractions 3-10 as 250µl each, fractions 11-14 as 500µl each.
- xvii. Also remove 20µl aliquots from each fraction in separate eppendorf tubes for quantification.
- xviii. Store the fractions and their respective aliquots at -80°C until further analysis or use.

1.3. Determination of virus concentration using agarose gel electrophoresis

- xix. Virus concentration can be determined by calculating viral genome equivalence over a 0.6% agarose gel in 1X TAE followed by staining with SYBR green.
- xx. Release viral DNA by denaturing the viral capsid by adding 10µl virus lysis buffer (0.1% Proteinase K, 0.5% SDS, 25mM EDTA in ddH₂O) to 10µl virus aliquot and incubate at 56°C for thirty minutes.
- xxi. Load entire volume and resolve via gel electrophoresis running known DNA concentration standards in parallel.

References:

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